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|---|-----|---|--------------------|-------------|
| 8 | (a) | packet/quantum/discrete amount of energy
of electromagnetic radiation
<i>(allow 1 mark for 'packet of electromagnetic radiation')</i>
energy = Planck constant $\times$ frequency <i>(seen here or in b)</i> | M1
A1

B1 |

[3] |
| | (b) | each (coloured) line corresponds to one wavelength/frequency
energy = Planck constant $\times$ frequency
implies specific energy change between energy levels
so discrete levels | B1

B1
A0 |

[2] |

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9 (a) (i) either probability of decay (of a nucleus)

M1